

Resources Now AI-Native Business Nervous System Latest Unified Product Requirement Document (PRD)

Resources Now is envisioned as an AI-native operational nervous system for businesses. It is designed to transform fragmented SMEs into intelligent, systemized, and self-coordinating organizations. The platform begins with the AEC (Architecture, Engineering & Construction) industry and later expands into all operational industries.

1. Vision & Philosophy

Resources Now is not merely ERP software, project management software, or a BIM tool. It is intended to become the operational infrastructure through which businesses think, coordinate, execute, and evolve. The platform acts similarly to a human nervous system: - AI Layer = Brain - Workflow Infrastructure = Spine - Department Connectivity = Nerves - Dashboards & Analytics = Eyes - Execution Engines = Muscles The software must adapt to the business instead of forcing businesses into rigid workflows.

2. Core Problem Statement

SMEs across industries suffer from fragmented software ecosystems, dependency on founders, disconnected departments, manual coordination, poor operational visibility, and reactive management styles. AEC firms specifically struggle with: - BIM coordination chaos - Version conflicts - Drawing management inefficiencies - Poor inter-department communication - Lack of centralized intelligence - Resource planning problems - Weak QA/QC systems Resources Now addresses these issues through a unified intelligent infrastructure.

3. Product Positioning

Resources Now should be positioned as: - A Business Operating System - Operational Intelligence Infrastructure - AI Coordination Layer - SME Transformation Platform - Enterprise Nervous System The platform should emotionally resonate with business owners by promising: - Systematic organization - Reduced chaos - Better visibility - Intelligent decision support - Entrepreneurial empowerment

4. Product Architecture Overview

The system architecture should be modular, scalable, AI-native, and highly customizable. The product architecture consists of: 1. Core Platform Layer 2. Workflow & Coordination Engine 3. AI Cognitive Layer 4. Industry Adaptation Layer 5. Integration Layer 6. Analytics & Dashboard Layer

5. Frontend Architecture

Frontend should behave more like a mission control center rather than a traditional dashboard. Recommended Stack: - React / Next.js - TypeScript - TailwindCSS - Zustand or Redux - WebSocket-based real-time updates - GraphQL for dynamic querying Frontend Modules: - Executive Dashboard - Department Cockpit - BIM Coordination Interface - Resource Allocation Center - AI Insights Panel - Task & Workflow Boards - Financial Overview Panels Key Frontend Principles: - Modular widgets - Adaptive layouts - Role-based UI rendering - Real-time operational visibility - AI-first interactions - Natural language query systems UX Philosophy: - Minimal clutter - Enterprise-grade clarity - Data hierarchy - Cognitive assistance - Human decision augmentation

6. Backend Architecture

Backend architecture must support enterprise scalability and modular customization. Recommended Backend Stack: - Node.js + NestJS - Python microservices for AI systems - PostgreSQL - Redis - Kafka / RabbitMQ - GraphQL Gateway - Docker + Kubernetes - AWS / Azure cloud infrastructure Backend Layers: 1. Authentication & Identity Layer 2. Workflow Engine 3. Business Logic Engine 4. AI Intelligence Layer 5. Notification System 6. Analytics Engine 7. Integration Gateway Microservices Recommended: - HR Service - Finance Service - BIM Coordination Service - Task Engine - AI Recommendation Service - Reporting Service - Audit & Logging Service

7. AI Cognitive Layer

This becomes the brain of Resources Now. AI Responsibilities: - Detect workflow bottlenecks - Predict delays - Recommend actions - Correlate operational patterns - Analyze resource utilization - Generate executive summaries - Forecast operational risks AI Concepts: - Agentic AI - Organizational Memory - Operational Reasoning - Decision Graph Systems - AI Workflow Agents - Predictive Coordination Potential AI Technologies: - LLM-based orchestration - Vector databases - Retrieval-Augmented Generation (RAG) - Business Knowledge Graphs - Time-series forecasting models

8. Customization Engine

Customization is one of the strongest differentiators. Businesses should be able to customize: - workflows - department structures - approval chains - dashboards - automations - hierarchy systems - notifications - operational pipelines Implementation Strategy: - No-code workflow builder - Dynamic schema engine - Rule-based automation systems - Plugin architecture - Industry template marketplace

9. AEC Industry Strategy

AEC should be the first industry because: - High operational complexity - Coordination-heavy workflows - Existing software fragmentation - BIM dependency - Document-heavy operations - Multi-stakeholder environments AEC Modules: - BIM Coordination - Drawing Management - Clash Detection Tracking - Site Progress Monitoring - RFI Systems - BOQ Systems - Consultant Coordination - QA/QC

10. Dashboard Philosophy

The dashboard should not behave like static reporting software. Instead it should: - infer meaning - prioritize issues - identify risks - recommend actions - simulate impacts - guide decision making The dashboard should feel like: - Mission control center - Business cockpit - Executive intelligence system - Operational brain interface

11. Monetization Strategy

Possible revenue streams: - SaaS subscriptions - Industry modules - AI usage pricing - Enterprise implementation - Marketplace ecosystem - Workflow templates - API ecosystem licensing Pricing should focus on: - modular adoption - scalable SME onboarding - low-friction entry

12. Technical Roadmap

Phase 1: - Core infrastructure - Authentication - Workflow systems - Basic dashboards - Task engine Phase 2: - AEC modules - BIM integration - AI insights - Resource planning Phase 3: - AI operational intelligence - Predictive systems - Autonomous workflows - Cross-industry expansion Phase 4: - Full ecosystem marketplace - AI operational agents - Enterprise-scale infrastructure

13. Biggest Risks

Key risks include: - Overengineering - ERP-level complexity - AI hallucinations - Poor onboarding experience - Excessive customization chaos - Slow SME adoption - Integration complexity Mitigation Strategies: - Start narrow - Focus on operational clarity - Build modularly - Avoid feature bloat - Prioritize usability

14. Long-Term Vision

Resources Now aims to become: - The operating system for SMEs - AI-native business infrastructure - Enterprise operational intelligence layer - Cross-industry execution platform At scale, businesses should rely on Resources Now as: - their operational memory - decision intelligence layer - execution nervous system - coordination environment

15. Final Strategic Insight

Most software helps businesses manage work. Resources Now aims to help businesses THINK. That philosophical shift is what differentiates it from traditional ERP systems, dashboards, project management tools, and business analytics software. The long-term success of Resources Now depends on: - exceptional UX simplicity - intelligent AI orchestration - deep operational understanding - modular customization - scalable architecture - strong execution discipline